NAME

nabla − find the longest common subsequence of two files

SYNOPSIS

nabla [*options*] [--] *file1 file2*

DESCRIPTION

nabla computes the Longest Common Subsequence (LCS) of *file1* and *file2*. By default it prints their differences as an edit script; with **-C** it prints the common (matching) records instead, and with **-l** it prints only the LCS length.

Each difference is introduced by a header line:

<<<<< n *file1* [*begin*:*end*] <<<<<
(records present only in *file1*)
>>>>> n *file2* [*begin*:*end*] >>>>>
(records present only in *file2*)

where *n* is the one-based difference number and *[begin:end]* the half-open range of records involved. A final line reports the LCS length, and when compiled with duration reporting enabled, a timing line such as *RickSpace: 28 operations in 9 µs @3.111 MHz* follows.

The two files are compared at one of three granularities:

record each input line is one record (the default);

text (**-T**) the whole file is one sequence of Unicode code points, so multi-byte characters are never split;

word (**-w**) each whitespace-separated word is one record.

The **-A** option specifies one of four LCS algorithms, or automatic dispatch, as follows:

a Automatic (the default): heuristic dispatch that selects the algorithm from the input's edit distance, match count, and alphabet size.

d Myers (1986): the O(ND) minimal edit-script algorithm (**N** the sum of the two input lengths, *D* the edit distance); fastest on near-identical input.

m Hunt-Szymanski (1977): a threshold array over the match lists, O((r+n) log n) time.

s RickSpace (Rick 2000): forward/backward contours plus Hirschberg recursion, O(ns) space.

t RickTime (Rick 1995): a single diagonal sweep, O(ns + d) space.

Automatic dispatch selects Myers when the edit distance is small, Hunt-Szymanski when the matches are sparse, RickTime when its match recording fits in memory, and RickSpace otherwise.

ARCHITECTURAL VARIANTS

nabla The default 32-bit position variant. It handles inputs of up to 4GB, which includes the human genome (˜3.1 billion), and stores each position in 4 bytes for maximal cache locality.

nabla64
The 64-bit position variant. Use it when the length of one or both inputs exceeds nabla's 32-bit (4GB) limit.

Both variants accept the same options and produce identical results for inputs that fit the 32-bit range. Building from source, the default **make -C cli** produces **nabla**, and **make -C cli LENGTH_SIZE=64 BUILD=build64** produces **nabla64**.

OPTIONS

-A *a*|*d*|*m*|*s*|*t*
Specify the LCS algorithm: **a** Automatic (the default), **d** Myers (1986), **m** Hunt-Szymanski (1977), **s** RickSpace (2000), or **t** RickTime (1995). Accepts **-Ad** or **-A d**.

-b Ignore whitespace: collapse each run of whitespace to a single space before comparing, and expand the reported spans back to the original positions.

-C Print the correspondence (the common, matching records) instead of the differences.

-F Treat the input files as FASTA and strip header lines; implies **-T**.

-G Emit the (gapped) CIGAR alignment string; file1 is the reference and file2 the query. Incompatible with **-C**, **-u**, and **-y**.

-h Print a usage synopsis and exit.

-H Print host information (name, locale, processor, and memory) and exit.

-i Perform a case-insensitive comparison.

-j *join* Coalesce nearby differences that lie within *join* records of each other. In unified **-u** mode it also merges intervals within that distance.

-l Report only the LCS length; skip reconstructing the differences.

-p *prefix*
Print *prefix* lines of context before each difference.

-P Draw a progress bar to standard error while the LCS is computed. The bar is shown only when standard error is a terminal.

-Q Treat *file1* and *file2* as literal strings rather than filenames; implies **-T**.

-s *suffix*
Print *suffix* lines of context after each difference.

-t *tabsize*
Set the tab size used to expand tab characters in the parallel **-y** output (default 8).

-T Compare the whole file as a sequence of Unicode code points (the text path) rather than line-by-line records.

-u Emit a GNU unified diff patch of the two files (the Edit format) instead of the default differences view. Context lines come from **-p** (prefix) and **-s** (suffix), defaulting to 3 for whichever is not given explicitly. **-u** is incompatible with **-C**, **-F**, **-G**, **-l**, **-Q**, **-T**, and **-y**. **-w** with **-u** is not yet implemented.

-v Enable verbose output: print the LCS length, the start and finish wall-clock times, and a timing summary; by default only the differences are shown.

-V Print the version and exit.

-w Split each line into whitespace-separated words and compare word by word (record mode).

-W *width*
Set the per-side content width used by the parallel **-y** format (default 80, minimum 5). Content is truncated to this width, so the total output line is 1 + 2 x width characters.

-y Emit side-by-side (parallel) output, in the manner of **diff**(1).

The **-h**, **-H**, and **-V** switches are early-exit switches: each prints its report and terminates without comparing files.

A **--** argument ends the options; the remaining arguments are operands, so a file whose name begins with a hyphen can follow it.

ALGORITHM

The **-A** option chooses among four algorithms, each fastest in its own regime. Three independent properties of the input decide which regime applies:

Edit distance
D: insertions plus deletions of the minimal edit script.

Match count ρ
r: the number of matched position pairs; low for sparse matches, high for skewed or small alphabets.

Alphabet size
: *s*: number of distinct symbols, which sizes the Rick table builds.

The algorithms and their regimes:

d Myers (edit distance): fastest on near-identical input (small *D*). The O(ND) minimal edit-script algorithm (**N** the sum of the two input lengths); its reconstruction history grows with the square of *D*, so it is bounded by a memory-derived budget.

m Hunt-Szymanski (matches): fastest when the matches are sparse (low ρ, large alphabet), the regime its threshold array is built for.

s RickSpace (space): the linear-memory fallback; O(ns) space via Hirschberg recursion, so it never exhausts memory and completes when no other algorithm does.

t RickTime (time): fastest when the matches are dense (high ρ), when its dominant-match recording fits in memory.

Automatic dispatch (**-Aa**, the default) estimates *D* and *r* from the symbol counts and selects in order:

1. Myers when the edit distance lower bound is at most 2000;
2. Hunt-Szymanski when the match count is at most 100 times the shorter input length (sparse matches);
3. RickTime when its recording budget covers the shorter input;
4. RickSpace otherwise.

-v reports which algorithm automatic dispatch selected and why.

EXIT STATUS

0 Success (including **-h**, **-V**, and **-H**).

1 Error: a command-line parse failure, a missing file, or an algorithm failure.

ENVIRONMENT

The locale (in particular **LANG**, **LC_ALL**, and **LC_NUMERIC**) controls the thousands separator used in the reported counts and the locale name shown by **-H**. On a system whose locale defines digit grouping, counts are printed with comma separators (for example **LCS Length = 4,591,960**); otherwise they are printed ungrouped.

EXAMPLES

Compare two files line by line and show the differences:

```
nabla old.txt new.txt
```

Compare side by side (parallel output) with 40-character sides and 2-space tab stops:

```
nabla -yp -W 40 -t 2 old.txt new.txt
```

Compare as text (code points), case-insensitively, showing only the length:

```
nabla -i -l -T a.fa b.fa
```

Compare two FASTA genomes with the linear-space algorithm and show matches:

```
nabla -As -C -F ecoli_mg1655.seq ecoli_mg1655_100subs.seq
```

Compare two literal strings rather than files:

```
nabla -Q "the quick sly fox jumped" "over the lazy brown dog"
```

SEE ALSO

diff(1), **cmp**(1)

The LCS algorithms and their performance bounds are described in *LCS Algorithms.md* and *LCS Performance.md* in the project documentation.